

Active Beauty

Vetivyne™

Fragrance inspired skin youth booster

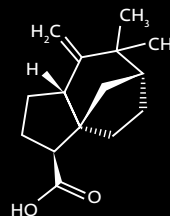
Crafted by green technology



Givaudan

engage your senses

Focus on the product



Zizanoic acid and dereplication of Vetiver root extract

Skin lipids and ageing

Skin lipids play a fundamental role in the maintenance of skin hydration, suppleness and barrier function. They are mostly produced by three types of cells:

- ▶ **Sebocytes**, located in the sebaceous glands, produce sebum, a thin lipidic film primarily composed of triglycerides, wax esters, squalene, and free fatty acids. Sebum ensures the skin protection from dehydration, the maintenance of its suppleness and also protects skin microflora by its acidic pH and its Antimicrobial Lipids (AML)¹.
- ▶ **Keratinocytes** accumulate lipids during their differentiation, such as ceramides and cholesterol derivatives, which will further build up the skin cornified envelope of the *stratum corneum*². This process is a key step in the preservation of skin barrier function and also prevents skin dehydration.
- ▶ **Adipocytes**, present in the subcutaneous fat, produce other types of triglycerides for energy storing, mechanical protection, and thermoregulation. They play an important role in skin volume, tonicity and firmness by ensuring mechanical support.

The disruption of the *stratum corneum* integrity usually starts with lipids depletion, with an overall decrease of -30% with age³. Lipids produced by the sebocytes are also affected, with a gradual decrease of sebum production around 25% per decade⁴. This results in a **loss of hydration of the skin**.

Meanwhile, facial fat loss upon ageing is related to a lower differentiation of the adipocytes⁵, and a decreased fat storage, resulting in **less firmness and tonicity, sagging and wrinkles appearance**.

Upcycling vetiver roots for skin beauty



Vetiver has been a source of inspiration for some of the most iconic fine fragrances in Perfume history. The most exquisite vetiver in the world is grown on the island of Haiti.

Since 2012 Givaudan has been collaborating with a local supplier to support a cooperative of vetiver root producers, who have achieved Ecocert Organic Standard and fair trade certification (ESR standard). This initiative ensures the social and environmental responsibility of vetiver production, enabling our perfumers to craft the most inspiring perfumes from a fully responsibly-sourced vetiver oil.

With **state-of-the-art extraction and purification processes**, Active Beauty experts have been able to re-use exhausted vetiver roots to create a fully natural, concentrated and odour-free extract showing impressive skincare benefits: Vetivyné™. Exclusive and patent-pending, rich in zizanoic acid, it has unique properties to reactivate the entire lipids synthesis in the skin.

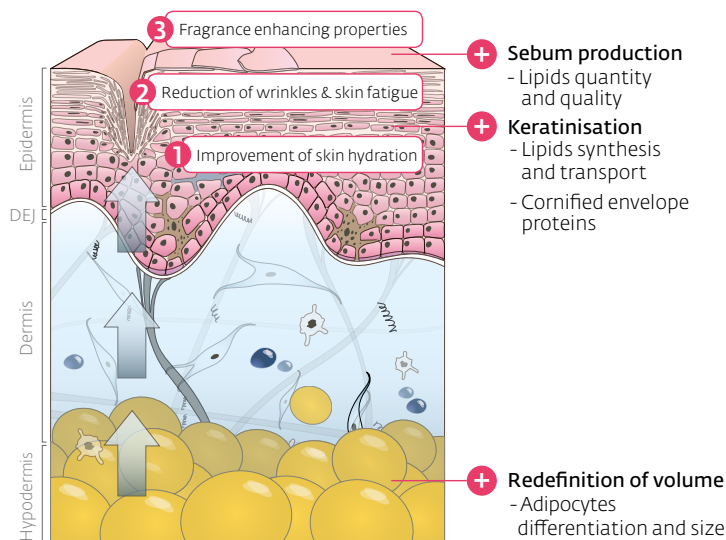
Vetivyné™ is a unique upcycled ingredient, using a side stream of Vetiver oil extraction to craft a high performance anti-ageing active.

Vetivyné™: mode of action

Vetivyné™ acts on the three main lipids sources in the skin, by improving the sebum production, the keratinisation and the adipocytes capacity to store fat.

Three clinical tests *versus* placebo have highlighted the consumers benefits of Vetivyné™:

1. Improvement of **skin hydration**
2. **Decrease of skin fatigue** and **reduction of wrinkles**, specifically in the perilabial area
3. **Fragrance enhancing properties**



¹ Fischer et al. 2014

² Feingold et al. 2014

³ Rogers et al. 1996

⁴ Downing et al. 1986

⁵ Tchokonia et al. 2010, Coleman et al. 2009, Kirland et al. 1997

Biological activity

Increasing lipids production for barrier function (*ex vivo*)

A proteomic analysis was performed by LC-MS/MS on human skin explants of 3 donors (54, 56 and 69 years old women) on which product was applied at 1%, everyday for 6 days.

Cellular function	Proteins name	Fold expression
Lipids metabolism & transport	ORM1 - Alpha 1 acid glyco-protein 1	0.57*
	RTN4 - Reticulon-4	1.33*
	CLTA - Clathrin light chain A	3.77*
	SNX5 - Sorting nexin 5	1.72*
	CERT - Ceramide transport protein	2.69*
	TFG - Protein TFG	1.34**
Epidermis differentiation	INV - Involucrin	1.59
	KRT36 - Keratin type I cuticular Ha6	1.73**
	DMKN - Dermokine	1.53**
	LOR - Loricrin	3.39*

Results: Vetivyn[™] stimulates the expression of a complete set of proteins involved in the multiple pathways targeting lipids production, lipids transport, and epidermis differentiation.

*p<0.05 Student's t-test **p<0.01 Student's t-test

Boosting sebum quantity and quality

1. Sebocytes stimulation (*in vitro*)

Sebocytes were pre-incubated with or without Vetivyn[™] at 1% for 4 hours. Then, a lipogenic mix (vitamin C, vitamin D3, insulin and calcium) was added for 7 days more of incubation. Treatment was renewed after 3 days. The lipid content was then evaluated using Bodipy[®] fluorescent labeling (lipids were made visible in green).

Results: Vetivyn[™] shows a significant stimulation of sebocytes, with an increase of sebum lipids accumulation up to +31% versus lipogenic mix only.

***p<0.001 Student's t test

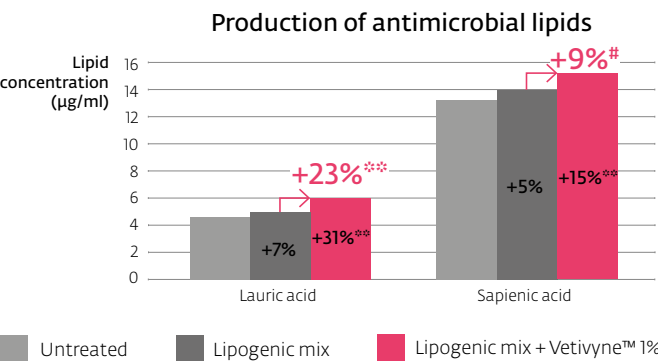
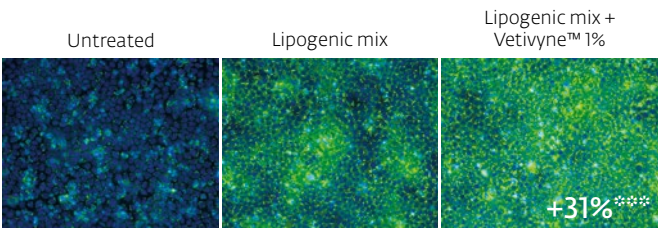
2. Antimicrobial lipids (AML) content increase (*in vitro*)

Sebocytes were cultivated in 3D culture and treated in systemic with or without Vetivyn at 1% for 7 days. The content of free fatty acids was then assessed by GC/MS to analyse their content in AML.

Results: Vetivyn[™] significantly increases up to +23% the content of antimicrobial lipids in sebum (Lauric acid and Sapienic Acid).

**p<0.01 Student's t-test #p<0.05 Student's t-test

Lipids accumulation in sebocytes



Biological activity

Barrier function reconstruction (*in vitro*)

1. Reactivation of lipids production (*in vitro* / *ex vivo*)

1.1 Ceramides and their precursors synthesis

Reconstructed Human Epidermis (RHE) were incubated with or without Vetivyné™ at 1% for 7 days. Lipids contained in RHE were then extracted and studied by Thin Layer Chromatography and densitometric analysis to assess the quantity of ceramides precursors in the RHE.

Results: Vetivyné™ significantly increases up to +42% the production of the ceramides and their various precursors in the epidermis.

**p<0.01 Student's t-test

1.2 Ceramides transport stimulation

Normal Human Keratinocytes (NHEKs) were stimulated for 5 days with or without Vetivyné™ at 1%. Ceramide Transport Protein (CERT) was then quantified by immunofluorescence.

Results: Vetivyné™ significantly increases up to +124.5% the expression of CERT, enabling a better transport of the lipids in the epidermis.

***p<0.001 Student's t-test

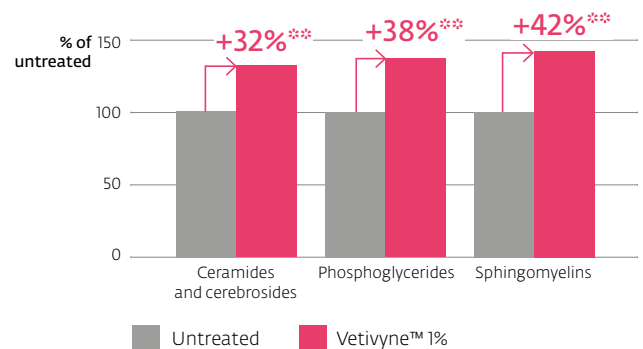
1.3 Boosting lipids content

Skin explants from a donor aged 69 were topically treated with or without Vetivyné™ at 1% for 6 days. The neutral lipids from skin were stained using LipidTOX™. The immunostaining was quantified by microscopical observation (green fluorescence) to assess stimulation of keratinisation.

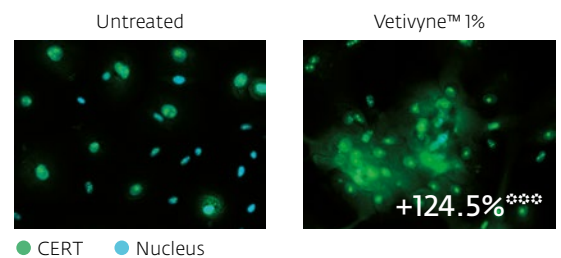
Results: As a result of boosting lipids production and transport, Vetivyné™ significantly increases up to +29.6% the lipid content in the cornified envelope.

**p<0.01 Student's t-test

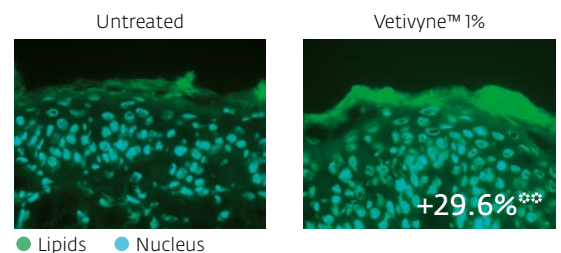
Ceramides and their precursors synthesis



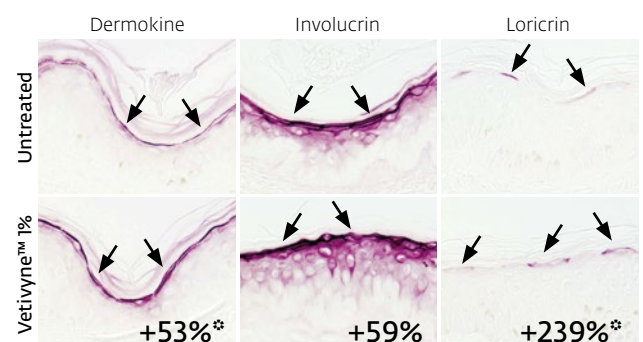
Ceramides transport stimulation



Boosting lipids content in cornified envelope



Increase of cornified envelope proteins (Illustrative pictures from explants of a 69 years old donor)



2. Reactivation of skin barrier proteins (*ex vivo*)

Human skin explants from 3 donors (54, 56 and 69 years old women) on which product was applied everyday for 6 days were assessed by proteomic using LC-MS/MS for quantification of dermokine, involucrin and loricrin, 3 characteristic proteins of the cornified envelope.

Results: Vetivyné™ at 1% significantly increases the expression of 3 major proteins of the *stratum corneum*, from +53% up to +239% in average.

*p<0.05 Student's t-test

Biological activity

Boosting skin adipocytes differentiation and size

1. Improving adipocytes differentiation (*in vitro*)

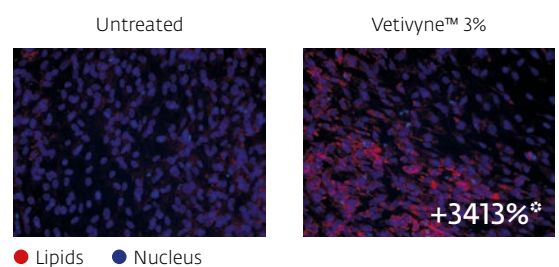
Pre-adipocytes were cultured for 13 days with or without Vetivyné™ at 3%.

Lipid droplets, a marker of adipocytes differentiation, were then labelled using AdipoRed™ (a fluorescent lipid probe) to quantify the differentiation.

Results: Vetivyné™ significantly **increases the adipocytes differentiation, with an effect up to +3413%***.

*p<0.05 Student's t-test

Stimulation of adipocytes differentiation

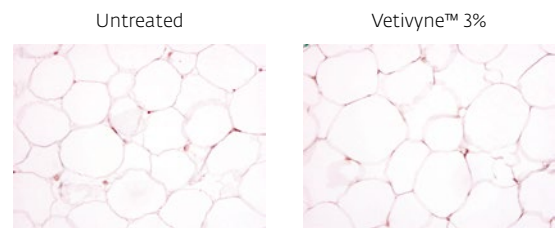


2. Increasing adipocytes volume (*ex vivo*)

Explants of full skin (epidermis, dermis, and hypodermis) were treated or not with Vetivyné™ at 3%. On D0, D1, D4 and D6, the product was topically applied. On D8, explants were stained for microscopical analysis.

Adipocytes size determination was then performed by image analysis, by measuring their equivalent circular diameter.

Increase of adipocytes size



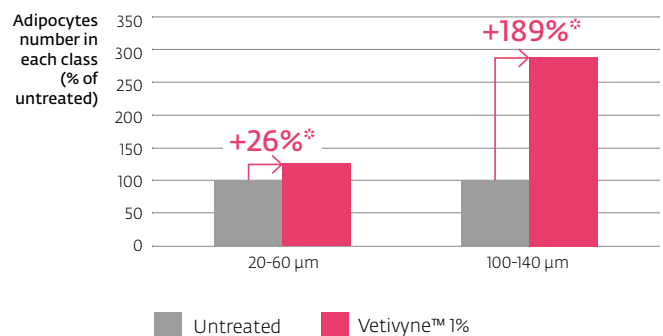
Results: Vetivyné™ significantly **increases the average size of adipocytes in the hypodermis**, as observed by microscopy.

It is also noticeable that a shift in the population of adipocytes can be observed after the use of Vetivyné™:

- ▶ The number of « small » adipocytes (20-60 µm) **increases by +26%, reflecting the differentiation boost observed *in vitro*** (appearance of new small adipocytes from pre-adipocytes)
- ▶ The number of « large » adipocytes (100-140 µm) **increases by +189%, as a consequence of the average size increase and a better fat storage capacity** (accumulation of lipids in the existing adipocytes)

*p<0.05 Student's t-test

Evolution of adipocytes size distribution and populations



Clinical efficacy

Visible skin fatigue reduction and decrease of perilabial wrinkles (Clinical test #1)

A double blind clinical evaluation was performed on 2 groups of 21 volunteers with sagging face and wrinkles (women from 52 to 69 years old, average of 60). Volunteers applied a placebo or the product containing Vetivyne™ at 2% on their face for 56 days, twice a day (morning and evening).

1. Improvement of tonicity/firmness and anti-fatigue effect

At D0, D28 and D56, skin biomechanical properties were measured by cutometry on cheekbones, focusing on the -R6 and R9 parameters, respectively representative of the viscoelastic balance and of the fatigue effect. At D0, D28 and D56, face of the volunteers was analysed thanks to AEVA HE®, a patented projection unit combined with stereo imaging.

Results: Vetivyne™ enables a time-progressive and significant recovery of the skin biomechanical properties on aged volunteers, with up to +11.2% recovery in tonicity and firmness and -17.8% of skin fatigue decrease versus placebo. These results are clearly visible on the 3D face images of the volunteers.

*p<0.05 Student's t-test

2. Smoothing of perilabial wrinkles

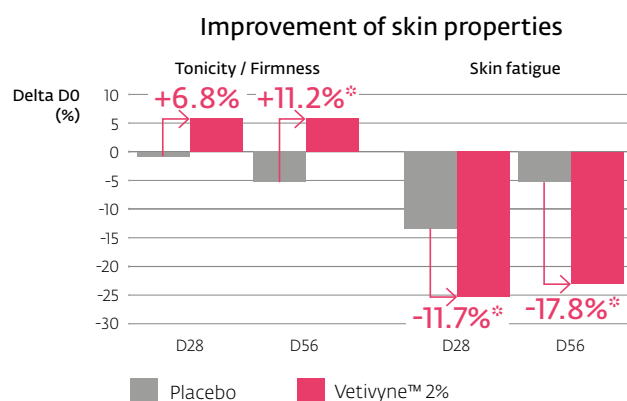
At D0 and D56 a particular focus was put into the perilabial wrinkles of the volunteers to quantify them thanks to AEVA HE®.

Results: The daily application of Vetivyne™ enables a significant effect for 100% of the volunteers versus placebo in 2 months, with an average reduction of -18%*** of the perilabial wrinkles.

***p<0.001 Student's t-test

3. Self-assessment of the skin benefits

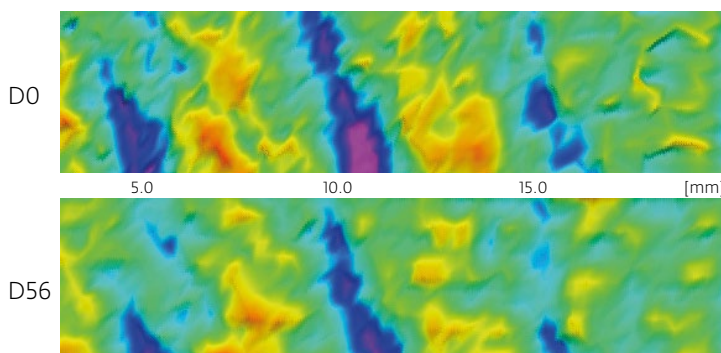
All volunteers were asked to assess the benefits of Vetivyne™ for their skin at D56.



3D face images
(Illustrative pictures from a 58 years old donor)



Decrease of perilabial wrinkles
(Illustrative pictures from a 60 years old donor)



Clinical efficacy

Skin composition and hydration benefits (Clinical test #2)

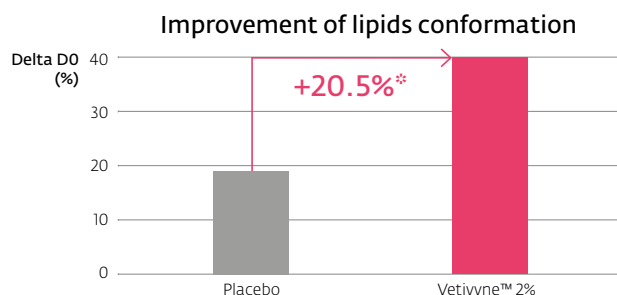
A double blind clinical evaluation was performed on 20 volunteers (women from 50 to 70 years old, average of 63), with dry legs (corneometry value <35). Volunteers applied the product containing either Vetivyné™ at 2% or a placebo on their legs twice a day (morning and evening) for 28 days.

1. Improvement of lipids conformation

At D0 and D28, Raman spectroscopy was used to assess the quality of lipids conformation in the skin.

Results: Vetivyné™ shows a significant **improvement of the lipids conformation in the skin, up to +20.5%**, therefore increasing their compacity, and ultimately the barrier function.

*p<0.05 Student's t-test

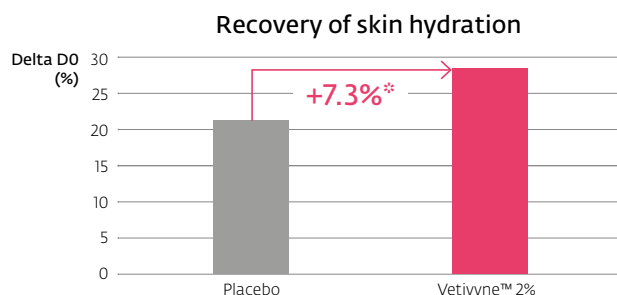


2. Increase of skin hydration

At D0 and D28, skin hydration was analysed by corneometry.

Results: Vetivyné™ shows a significant **increase of skin hydration up to +7.3%** on volunteers with dry skin.

*p<0.05 Student's t-test



Fragrance boosting benefits (Clinical test #3)

Thanks to the observed skin benefits (lipids quantity and conformation improvement), hypothesis was made that Vetivyné™ could impact on fragrances behaviour by modifying the skin barrier properties.

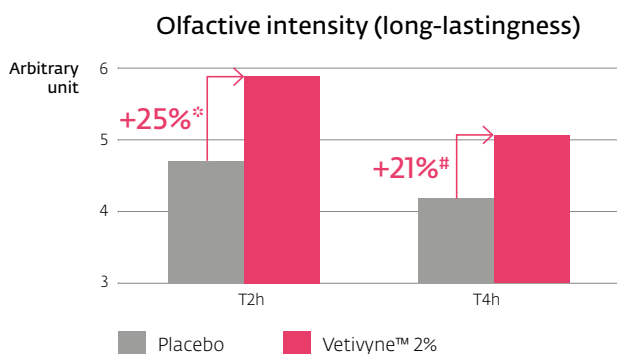
To assess the long-lastingness of a fragrance, a clinical test was performed on 20 women. Volunteers applied twice a day a placebo cream on one of their forearm and a cream containing Vetivyné™ at 2% on the other for 1 month. Then, on the last day, a fine fragrance was applied to both of their forearms.

1. Self-assessment of long-lastingness

Volunteers were asked to rank the olfactive intensity of the fine fragrance 2 hours and 4 hours after its application on their skin.

Results: Vetivyné™ demonstrates a **boosting effect of fragrances long-lastingness** by significantly **increasing the fragrance olfactive intensity hours after application, up to +25%**.

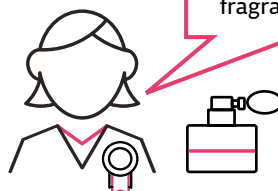
*p<0.05 Student's t-test #p<0.1 Student's t-test



By emphasising fragrances heart and base notes over time, Vetivyné™ demonstrates a real interest as a fragrance sensuality booster.

2. Professional fragrance evaluator opinion

To confirm the benefits of Vetivyné™ in terms of fragrance boosting properties, a professional fragrance evaluator was asked to assess the differences in terms of olfactive properties between the 2 forearms of the volunteers.



Summary



Approval pending

Technical information

Proposed INCI:	Vetiveria Zizanoides Root Extract (and) Propanediol (and) Water
Origin:	Vegetal extraction
Preservation:	Preservative free
Appearance:	Amber liquid
Solubility:	Water soluble
Dosage:	1-3%
Processing:	Can be added at the beginning or at the end of the formulation process into the water phase. Formulate at temperature below 40°C, and pH between 4 and 10.

Claims

Claims:	Stimulation of sebum production, stimulation of sebum antimicrobial lipids production, activation of adipocytes volume increase, skin hydration, skin tonicity booster, skin fatigue reduction, perioral wrinkles reduction, skin repulping, fragrance long-lastingness enhancer.
Applications:	Serums for dry skin, anti ageing perioral serum, anti -ageing night and day creams, anti fatigue essence, skin tonifying mask, anti-acne cream, body lotion to increase fragrance long-lastingness.

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